Power Electronics Laboratory
Session 3: Isolated DC-DC Converters
(Flyback and Forward Converters)

Objectives

1. Understand the operation of the flyback converter.
2. Understand the operation of the forward converter.
3. Evaluate the role of transformer and reset mechanisms.

General instructions

1. Introduction is evaluated with **0.5** points maximum.
2. Conclusion is evaluated with **0.5** points maximum.
3. For every **3** figure format and order error, **0.1** points will be discounted.
4. All figures and equations must be referenced.
5. The student must decide what variables to plot when not explicitly indicated.

Part 1: Flyback Converter

Build the flyback converter shown in Fig. 1 using Plecs.

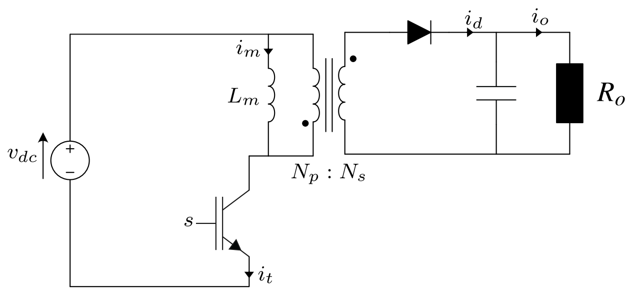

Figure 1: Flyback converter.

Use the following parameters:

- $V_{dc} = 100$V
- $f_s = 10$kHz
- $R = 4\Omega$
- $N_p = 1$, $N_s = 1$
- $L_m = 1mH$
- $C_o = 1000\mu F$
- Transistor on resistance $= 1m\Omega$

Note: There is an option to change the polarity of the transformer winding, use +- to have positive and negative polarity in primary and secondary windings.

Task 1: Continuous Conduction Mode (CCM)

1. Select **three** duty cycles to obtain output voltage below, equal, and above the input voltage. For all duty cycles, the converter must operate in CCM in steady state.
2. Simulate until steady-state operation is reached.
3. Plot at least 3 switching cycles of relevant variables.
4. Verify that the converter operates in CCM.
5. Analyze and comment the following:
   - Output voltage
   - Input current
   - Magnetizing current waveform and ripple

Task 2: Discontinuous Conduction Mode (DCM)

1. Choose one duty cyle and apply a load step to Increase the load resistance (Decrease the load) to force DCM operation.
2. Simulate steady state.
3. Plot relevant waveforms.
4. Identify the instant when the magnetizing current reaches zero.
5. Compare CCM and DCM operation for the same duty cycle:
   - Output voltage variation
   - Current waveforms

Part 2: Forward Converter

Build the forward converter shown in Fig. 2.

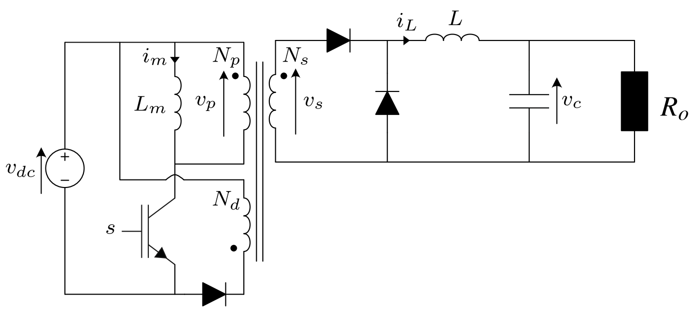

Figure 2: Forward converter.

Use the following parameters:

- $V_{dc} = 100\text{V}$
- $f_s = 10\text{kHz}$
- $R_o = 4\Omega$
- $N_p = 1$, $N_d = 0.1$, $N_s = 2$
- $L = 1mH$
- $C_o = 1000\mu F$

- Transistor on resistance = $1m\Omega$

Note: Apply the correct winding polarities.

Tasks: Duty Cycle Analysis

1. Select **three** duty cycles. Check that the converter always operate in CCM and that the reset circuit operate correctly.
2. Simulate steady-state operation.
3. Plot relevant variables.
4. Analyze:
   - Output voltage
   - Inductor current
   - Magnetizing current

Tasks: Reset Circuit Analysis

1. Choose a duty cycle in which the system operate in CCM and the reset circuit works properly.
2. Apply a duty cycle step change to increase the duty cycle. The new duty cycle should be chosen in such manner that the reset circuit stops working properly (The magnetizing current of the transformer increases in each switching period).
3. Analyze and justify why the reset circuit stops working with the new duty cycle (Current derivative, Off time)